

ESG Intelligence Briefing — July 28, 2026

Executive Summary

The full implementation of the EU Carbon Border Adjustment Mechanism (CBAM) in July 2026 presents an immediate and material challenge for Kazakhstan's steel and cement exporters, mandating carbon reporting and potentially imposing significant levies. Concurrently, the global construction market is witnessing a profound shift towards demanding verifiable low-carbon materials, driven by tightening procurement rules, increased investor scrutiny, and a preference for Environmental Product Declarations (EPDs). Clients must prioritize carbon transparency and explore low-carbon material innovations to maintain market access and competitiveness.

What Changed Since Last Briefing

Since the last briefing on July 12, 2026, the focus on the impending full CBAM implementation for specific materials like steel and cement has intensified, directly impacting Kazakhstan. The market also shows an accelerated trend in the demand for robust carbon data via EPDs and a shift towards performance-based material standards to drive decarbonization, notably in concrete and steel production pathways (e.g., green hydrogen). The last briefing highlighted CBAM impacts and low-carbon concrete trends, and these themes are now gaining critical momentum.

1. Regulation Watch

- **EU Carbon Border Adjustment Mechanism (CBAM) Full Implementation:** The transitional period for CBAM is ending in July 2026, moving to full implementation. This directly impacts imports of electricity, aluminium, iron and steel, and cement into the EU. Kazakhstan's exporters of steel and cement will now face a carbon levy based on embedded emissions, requiring detailed reporting. This aims to level the playing field for European producers and encourages global industrial decarbonization.
- **Who it affects:** Kazakhstan's steel and cement exporters, EU importers, and indirectly, all construction entities sourcing from or operating within the EU supply chain.
- **Effective Date:** July 2026 (full implementation).
- **Business Implication:** Kazakhstani exporters must establish robust systems for measuring and reporting embodied carbon in their products to avoid significant carbon levies and potential competitive disadvantage. Investment in decarbonization of production processes is critical.
- **Source:** *Web search summary (Assessment of impacts of the EU-CBAM)*, *ICAP Carbon Action (<https://icapcarbonaction.com/en/news/eu-carbon-border-adjustment-mechanism-cbam-takes-effect-transitional-phase>)*
- **Tightening Procurement Rules in EU and UK:** Demand for greener materials is no longer aspirational but is now shaping government and major buyer procurement frameworks. These frameworks reward transparency, prioritize low-emissions performance, and make carbon data essential. The trend is moving from encouragement to expectation, with a clear direction that carbon intensity is becoming a key competitive edge.
- **Who it affects:** All construction material suppliers and contractors bidding on public and large-scale private projects in the EU and UK, and potentially other markets adopting similar standards.
- **Effective Date:** From 2026 onwards.
- **Business Implication:** Clients must ensure their supply chains can provide verified carbon data and demonstrate lower-emissions performance to secure future contracts.

- **Source:** *The Climate Group
(<https://www.theclimategroup.org/our-work/news/2026-year-rules-are-changing-steel-and-concrete>)*

2. Construction & Market News

- **Performance-Based Standards for Concrete:** A significant shift is underway in concrete standards, moving from prescriptive ingredient lists to performance-based approaches. This prioritizes strength, durability, and overall performance, opening the door for innovation in low-carbon concrete mixes, including the use of AI-enabled optimization. This change is gaining ground as building codes evolve beyond legacy rules.
- **Source:** *The Climate Group
(<https://www.theclimategroup.org/our-work/news/2026-year-rules-are-changing-steel-and-concrete>)*
- **Green Hydrogen Steel Production Scaling:** Green hydrogen-based Direct Reduced Iron (DRI) is moving beyond pilot projects towards commercial production, with significant deals (e.g., Stegra's agreement with Thyssenkrupp) and progress towards commercial hydrogen-reduced steel from 2026 (e.g., SSAB). This signifies growing commercial confidence in near-zero primary steel production pathways.
- **Source:** *The Climate Group
(<https://www.climategroup.org/our-work/news/2026-year-rules-are-changing-steel-and-concrete>)*
- **Investor Scrutiny Driving Low-Carbon Material Uptake:** The private sector, particularly developers and building owners, is increasingly weaving embodied carbon into material choices. This is driven by investor scrutiny, internal sustainability targets, and the need to future-proof assets. Data centers are leading this trend, with their scale and long horizons encouraging early adoption of low-carbon concrete.
- **Source:** *The Climate Group
(<https://www.theclimategroup.org/our-work/news/2026-year-rules-are-changing-steel-and-concrete>)*

3. ESG Disclosure Signals

- **EPDs as a Business Baseline and Competitive Asset:** From 2026, carbon accounting, particularly through Environmental Product Declarations (EPDs), is increasingly shaping a company's ability to compete. 90% of AEC professionals prefer products with EPDs, and 65% require them for all or most projects. EPDs are seen as critical for achieving up to 30% embodied carbon reductions and are becoming a strategic, competitive asset for manufacturers.
- **Signal:** High demand for EPDs indicates a market-wide shift towards data-driven sustainability decisions and transparency. Manufacturers without EPDs risk market access and competitive disadvantage.
- **Source:** *One Click LCA "2026 Carbon Experts Report"
(<https://oneclicklca.com/resources/research/2026-carbon-experts-report>)*
- **Automation and AI for EPD Creation:** Automation and AI are widely recognized as key enablers for scaling, consistency, and speed in Life Cycle Assessments (LCAs) and EPD creation, especially when underlying data is robust. This indicates a growing trend towards digital solutions to overcome the cost and complexity manufacturers face in covering entire product portfolios with EPDs.
- **Signal:** The industry is looking for efficient ways to meet the surging demand for verifiable product carbon data.
- **Source:** *One Click LCA "2026 Carbon Experts Report"
(<https://oneclicklca.com/resources/research/2026-carbon-experts-report>)*

4. Embodied Carbon Spotlight

- **Automated LCA Scenario (Tool Limitation):** Due to technical limitations with the automated embodied carbon analysis tools, a specific cradle-to-gate (A1-A3) LCA on a representative construction scenario could

not be performed for this briefing. Therefore, no quantitative results table or specific material hotspots can be presented today.

- **Embodied Carbon Trends:** Despite the tool limitation, the market trends clearly indicate a strong push for embodied carbon reduction:
- **Material Decarbonization:** Significant efforts are underway to reduce the carbon footprint of essential construction materials like steel and concrete. This includes the scaling of green hydrogen-based steel production and the development of performance-based standards for concrete that encourage innovative, low-carbon mixes.
- **Data and Transparency:** The increasing demand for Environmental Product Declarations (EPDs) demonstrates a critical need for transparent and verifiable embodied carbon data. EPDs are becoming a de-facto requirement for market access and competitive bidding, with 90% of AEC professionals preferring products with them.
- **Early-Stage Impact:** The greatest potential for embodied carbon reduction lies in the early stages of design and pre-specification. Making informed material choices early can lead to significant savings.

Recommended Actions

1. **Assess CBAM Exposure Immediately:** Kazakhstan-based clients exporting steel, cement, or related products to the EU must conduct an urgent assessment of their embodied carbon emissions and establish robust reporting mechanisms to comply with CBAM and mitigate financial liabilities from July 2026.
2. **Prioritize EPD Development/Procurement:** Manufacturers should invest in developing EPDs for their key products, leveraging automation and AI where possible, to meet growing market demand and gain a competitive edge. Contractors and developers should prioritize specifying materials with readily available EPDs.
3. **Explore Low-Carbon Material Innovations:** Actively research and pilot projects utilizing low-carbon concrete (especially those adhering to performance-based standards) and assess the viability of procuring green hydrogen steel as it becomes commercially available.
4. **Integrate Embodied Carbon into Early Design:** Implement Life Cycle Assessment (LCA) tools and expertise at the earliest design stages to compare material options and lock in significant carbon reductions before construction begins.
5. **Educate and Train Teams:** Provide targeted training for procurement, design, and project management teams on embodied carbon, CBAM implications, and the value of EPDs to foster a data-driven sustainability culture.

Sources

- Web search summary on "EU CBAM transitional period updates July 2026 impact on Kazakhstan steel cement"
- International Carbon Action Partnership: *EU Carbon Border Adjustment Mechanism (CBAM) takes effect with transitional phase* (<https://icapcarbonaction.com/en/news/eu-carbon-border-adjustment-mechanism-cbam-takes-effect-transitional-phase>)
- The Climate Group: *2026 is the year that the rules are changing for steel and concrete* (<https://www.theclimategroup.org/our-work/news/2026-year-rules-are-changing-steel-and-concrete>)
- One Click LCA: *2026 Carbon Experts Report* (<https://oneclicklca.com/resources/research/2026-carbon-experts-report>)

Generated by the ESG Intelligence Agent on July 28, 2026. Embodied-carbon figures are indicative cradle-to-gate (A1-A3) estimates — verify with product-specific EPDs before formal reporting.